XRPL Validator — Daily Engineering Report · 2026-09-25

Period: 2026-09-25 00:00 to 24:00 US Eastern (2026-09-25 04:00 to 2026-09-26 04:00 UTC); ledgers #107,218,189 to #107,240,423. Every count from the ledger and the validator's logs stops at #107,240,423. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; signing records; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Normal (agreement not checked) — coverage outside planned maintenance 100.000% (objective 99.9%), agreement with the network not checked (the reference node no longer holds these ledgers' hashes), no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,211 of the 22,235 ledgers the network closed (99.892%) (agreement not checked: the ledgers' hashes are no longer held). Unsigned: 24 during planned restarts.
- **Native engine:** 4 engine receipts recorded (4 fixed, 0 explained, 0 without a recorded fix).
- **Changes:** 2 deployments (cycles 155, 156), 6 commits merged into main, 2 engine findings fixed (F401–F402).

Service levels

Measure	Value
Ledgers closed by the network	22,235
Validations issued	22,211
Coverage (validations / ledgers closed)	99.892%
Validations matching the network's validated ledger hash	not checked: the reference node no longer holds these ledgers' hashes
Unsigned during planned restarts	24
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%

Measure	Value
Restarts	3
Basis: the hash the first peer announced as accepted (before cycle 183)	22,211

Restarts:

- 13:47:05 UTC: deployment of cycle 155; validations resumed at #107,227,260 (8 ledgers unsigned between the last validation before the restart and the first after it)
- 14:53:31 UTC: deployment of cycle 156; validations resumed at #107,228,283 (9 ledgers unsigned between the last validation before the restart and the first after it)
- 15:33:20 UTC: restart (no deployment recorded); validations resumed at #107,228,896 (7 ledgers unsigned between the last validation before the restart and the first after it)

Other planned maintenance:

- **107,228,889–#107,228,895 (7): A deliberate warm restart at 15:33 UTC, one minute after the reference node finished its upgrade to rippled 3.4.1, to move the validator off the public WebSocket fallback it used during that upgrade.**

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,218,189–#107,240,423), by step:

Step	Ledgers with a receipt
Step 1 · native engine	4
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#107,219,029	Step 1 · native engine	1 transaction result(s) differed from rippled's; 9 ledger change(s) the network made were absent from ours: 8321932A, DB9E82ED, 81B0D5DC; 1 object(s) differed in content	fixed: F401, commit 166f033, deployed in cycle 155 (2026-09-25 13:47 UTC)
R-02	—	#107,227,378	Step 1 · native engine	1 transaction result(s) differed from rippled's; 10 ledger change(s) the network made were absent from ours: CAD7A38C, 50E963A9, 1610F05D; 2 object(s) differed in content	fixed: F402, commit e63a7d9, deployed in cycle 156 (2026-09-25 14:53 UTC)
R-03	—	#107,227,684	Step 1 · native engine	1 transaction result(s) differed from rippled's; 10 ledger change(s) the network made were absent from ours: 160B31C0, 2637E0DD, A9171EE6; 2 object(s) differed in content	fixed: F402 — a TicketCreate paid with a Ticket does not count the ticket it spends; one bot account; deployed in cycle 156 (2026-09-25 14:53 UTC)
R-04	—	#107,227,978	Step 1 · native engine	1 transaction result(s) differed from rippled's; 10 ledger change(s) the network made were absent from ours: B7880A10, 76D51EBB, 2C58AE1B; 2 object(s) differed in content	fixed: F402 — a TicketCreate paid with a Ticket does not count the ticket it spends; one bot account; deployed in cycle 156 (2026-09-25 14:53 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-25-01, INC-01 is INC-2026-09-25-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
155	`0c76af1`	13:47:05 UTC	n/a	n/a	400 matched, 0 mismatched	#30 from 14:14 UTC: ended early: superseded by the next deployment (2026-09-25 15:20 UTC)
156	`677c18d`	14:53:31 UTC	n/a	n/a	406 matched, 0 mismatched	#31 from 15:20 UTC: ended early: superseded by the next deployment (2026-09-26 17:21 UTC)

Cycle 155 delivered: `458c2c7` fix(node): validation relay answers the hubs' keep-alive pings; `1c373c9` fix(node): voting-ledger validations keep their fields in canonical order; `166f033` fix(ledger): F401 — AMM withdrawal reserve check counts XRP already paid out (soak #29 receipt); `e63a7d9` fix(ledger): F402 — TicketCreate paid with a Ticket does not count the ticket it spends; `0c76af1` feat(node): stop voting for SingleAssetVault and LendingProtocol

Cycle 156 delivered: `677c18d` test(node): campaign 25 — Batch x Clawback, AMMClawback and token escrow (devnet vectors)

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
04:18 UTC	xrpl-validator	`458c2c7`	fix(node): validation relay answers the hubs' keep-alive pings
04:40 UTC	xrpl-validator	`1c373c9`	fix(node): voting-ledger validations keep their fields in canonical order
05:10 UTC	xrpl-validator	`166f033`	fix(ledger): F401 — AMM withdrawal reserve check counts XRP already paid out (soak #29 receipt)
14:07 UTC	xrpl-validator	`e63a7d9`	fix(ledger): F402 — TicketCreate paid with a Ticket does not count the ticket it spends
14:08 UTC	xrpl-validator	`0c76af1`	feat(node): stop voting for SingleAssetVault and LendingProtocol
14:31 UTC	xrpl-validator	`677c18d`	test(node): campaign 25 — Batch x Clawback, AMMClawback and token escrow (devnet vectors)

Findings fixed

- **fix(ledger): F401 — AMM withdrawal reserve check counts XRP already paid out (soak #29 receipt)** (`166f033`, xrpl-validator) rippled's AMMWithdraw::withdraw checks the reserve for a new trust line against max(priorBalance, balance) (AMMWithdraw.cpp:705), where balance is the account's live balance. It pays the transaction's Asset before it checks Asset2, so when Asset is XRP the check sees the XRP already withdrawn. withdraw_reserve_ok compared the pre-fee balance alone, which is exact for a single-asset withdrawal but not for a two-asset …
- **fix(ledger): F402 — TicketCreate paid with a Ticket does not count the ticket it spends** (`e63a7d9`, xrpl-validator) rippled's Transactor::apply runs consumeSeqProxy before doApply: a transaction paid for with a Ticket deletes that ticket first, lowering OwnerCount and TicketCount by one. TicketCreate's checks see the account after that: doApply requires the pre-fee balance to cover accountReserve(OwnerCount + TicketCount), and preclaim refuses with tecDIR_FULL only when curTicketCount + addedTickets - consumedTickets exceeds 250. …

Other engineering changes

- **fix(node): validation relay answers the hubs' keep-alive pings** (`458c2c7`, xrpl-validator) live_viewer's validation relay (val-sender) read and discarded everything the relay hubs sent, including their keep-alive pings. rippled pings each peer on a 60-second timer and drops one that has not answered the previous ping by the next tick (PeerImp::onTimer, "Ping Timeout"), so every relay connection closed about two minutes after it opened, and the relay learned of it only when its next write failed. On …
- **fix(node): voting-ledger validations keep their fields in canonical order** (`1c373c9`, xrpl-validator) rippled deserializes a peer's validation with requireCanonicalOrder (PeerImp::onMessage(TMValidation) -> STObject::set; present in 3.3.0 and 3.4.0, absent in 3.1.2): each field's code must exceed the previous one, or the validation is dropped ("Fields in object are not in canonical order") and the sender is charged for invalid data. sign_validation appended the Signature (VL, field 6) after every other field. On the …
- **feat(node): stop voting for SingleAssetVault and LendingProtocol** (`0c76af1`, xrpl-validator) Either amendment switches rippled's Number to its 19-digit mantissa for every transaction: setCurrentTransactionRules (Rules.cpp) selects the Large mantissa scale when SingleAssetVault or LendingProtocol is enabled. The engine

models the 16-digit scale only, so this validator no longer votes for either. Decision of the validator's operator, 2026-09-25.

Tests and campaigns

- **test(node): campaign 25 — Batch x Clawback, AMMClawback and token escrow (devnet vectors)** (`677c18d`, xrpl-validator)

Independence program

No independence record for UTC day 2026-09-25; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Voting events:

- fixBatchV1_2 gained majority — ledger 107,227,649, 2026-09-25 14:13 UTC
- BatchV1_1 lost majority — ledger 107,227,905, 2026-09-25 14:29 UTC
- BatchV1_1 gained majority — ledger 107,228,161, 2026-09-25 14:46 UTC

Amendment	Majority since	Activation due
PermissionDelegationV1_1	2026-09-24 21:25 UTC	2026-10-08 21:25 UTC
fixBatchV1_2	2026-09-25 14:12 UTC	2026-10-09 14:12 UTC
BatchV1_1	2026-09-25 14:46 UTC	2026-10-09 14:46 UTC

Amendments enabled on mainnet: 94.

Open items

None.

Notes

Hand-written notes on the day's XRPL work: research and operations.

- **Deployment and the first warm restart.** Build 166f033 (findings F401, the voting-ledger validation order and the validation relay, all described below) replaced the live build at 09:46 EDT after its gate (1,636 suite tests and 707 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches). It was the first deployment to use the warm restart merged on 2026-09-24: the stopping build recorded its last verified ledger, and the new build rebuilt its stored state in 22.1 seconds, confirmed that the state's root equals the network's account hash for ledger 107,227,250, and resumed two ledgers behind the network. Cold starts had resumed 90–111 ledgers behind, several minutes after the switch. The health check matched 400 ledgers with no mismatches; soak #30 began at 10:14 EDT.
- **Voting-ledger validations.** Every 256th ledger (sequence ≡ 255 mod 256) is a voting ledger: a validator's validation of it carries its amendment votes. The node signed those validations but placed the signature after the amendment list, which puts the fields out of canonical order, and rippled drops a validation whose fields are out of order (the check is present in rippled 3.3.0 and 3.4.0). The reference node's record from 2026-09-24 12:48 to 2026-09-25 00:22 EDT has our validation for none of the 41 voting ledgers in that period, so the network never received our amendment votes; it has our validation for every other ledger after 21:00 on 2026-09-24. The signature now takes its canonical place; the signed content is unchanged. Before deployment this was verified on mainnet with a throwaway key: the same voting-ledger validation, sent through a public hub, reached the reference node 93 ms later in the corrected layout and did not arrive in the old one. After deployment the reference node received our validation of voting ledger 107,227,391 at 09:56 EDT, the first voting-ledger validation of ours in its record. Fixed in 1c373c9.
- **Amendment votes.** The validator no longer votes for SingleAssetVault or LendingProtocol. Either amendment switches rippled's `Number` arithmetic to a 19-digit mantissa for every transaction, and the engine models the 16-digit scale that mainnet uses. Deployed at 10:53 EDT (below); from voting ledger 107,228,415 the validator's votes cover 46 amendments instead of 48.
- **Validation relay.** Besides its regular peer connections, the node sends its validations over four dedicated connections to well-connected hub servers. Those connections read and discarded everything the hubs sent, including their keep-alive pings. rippled pings each peer on a 60-second timer and drops a peer that has not answered its previous ping by the next tick, so each relay connection closed about two minutes after it opened; on 2026-09-24 the four relays reconnected 87 times in one hour. These drops did not cost validations at the reference node, which received every non-voting validation after 21:00 through the other connections. The relays now answer the pings, notice a closed connection at once, and after reconnecting re-send the last validation if it went out less than 60 seconds earlier. Fixed in 458c2c7 and deployed at 09:46; from the warm restart at 11:33 to the end of the day, each of the three hubs that accept peer connections held a single relay session with no drop. The fourth hub no longer accepts peer connections and will be replaced.
- **Soak #29 receipt and finding F401.** At ledger 107,219,029 an AMMWithdraw (withdraw all, from an XRP/DROP pool) came from an account holding 2.310461 XRP at an owner count of 6 with no DROP trust line; a new line at that owner count needs a 2.4 XRP reserve. rippled checks that reserve against the larger of the balance before the fee and the account's current balance, and it pays the XRP side out before it checks the DROP side, so the check saw 4.753549 XRP and mainnet accepted the withdrawal. The check added for F399 on 2026-09-24 compared only the balance before the fee and refused it. Fixed in 166f033 (deployed above).
- **Finding F402.** At ledger 107,227,378, during the new build's health check, a TicketCreate of 10 tickets paid for with a Ticket came from an account holding 3.398367 XRP at an owner count of 2. rippled deletes the ticket being spent before TicketCreate checks its reserve, so ten more tickets needed 1 + 0.2 × 11 = 3.2 XRP and mainnet succeeded; the engine counted the spent ticket, required 3.4 XRP and refused. The same account repeated the

pattern at ledger 107,227,684. Fixed in e63a7d9, with the mainnet transaction pinned as a regression vector, and deployed at 10:53: the same account's next ticket-funded TicketCreate inside the affected balance range, at ledger 107,228,620, matched mainnet.

- **Campaign 25 (devnet).** Forty byte-exact vectors for Batch (BatchV1_1), combined with trust-line and MPT clawback, AMMClawback, and MPT and IOU escrow, all already enabled on mainnet. Thirty-seven are byte-exact under both flow engines, with no engine finding. The other three differ only because devnet enables SingleAssetVault and LendingProtocol and so runs `Number` at 19 digits; libxrpl 3.4.0 under devnet's rules reproduces devnet's results. Merged as 677c18d.
- **fixBatchV1_2.** An amendment named fixBatchV1_2 gained majority on mainnet at 14:12 UTC. The reference node's rippled 3.4.0 did not recognise it; rippled 3.4.1, which does, is available from the XRPL Foundation's package repository (package xrpld 3.4.1), and the reference node was upgraded to it the same day (below). Unless it loses majority, the amendment activates about 2026-10-09 14:12 UTC; the engine must implement it once its source is published. BatchV1_1 lost majority for one voting round (ledgers 107,227,905 to 107,228,161, 14:29 to 14:46 UTC), which restarted its two-week period: Batch now activates about 2026-10-09 14:46 UTC, about half an hour after fixBatchV1_2, instead of 2026-09-29.
- **Second deployment (F402 and the vote change).** Build 0c76af1 replaced the live build at 10:53 EDT after its gate (1,640 suite tests and 707 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches), again by warm restart: it rebuilt its state in 22.1 seconds and resumed one ledger behind the network. The health check matched 406 ledgers with no mismatches, and soak #31 began at 11:20 EDT. It recorded no receipts through the end of the day.
- **Reference node on rippled 3.4.1.** The reference node was upgraded from rippled 3.4.0 to 3.4.1 at 11:22 EDT and was fully synchronised again at 11:32, after re-acquiring the network's ledger; it recognises fixBatchV1_2. While it restarted, the validator switched to a public fallback server; a warm restart at 11:33 returned it to the reference node, 145 ledgers behind at resume and level with the network by 11:35, with no mismatches at any point. The engine's built-in copy of the rippled library stays at 3.4.0 until the 3.4.1 source is published; the two produce identical results until fixBatchV1_2 is enabled.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Attested Reserve Invariant for Wrapped Bridge Assets** (interop) Add sfAttestedLockedBalance to the Bridge ledger object and a new witness-quorum transaction XChainAttestReserve, reusing the door account's existing
- **Atomic Multi-Leg Escrow Finish for DvP** (settlement) Add sfLinkedEscrows, a bounded (max 4) STVector256 of Escrow keylets, to EscrowFinish, and extend the transactor so that when present, doApply verifie

- **Credential-Gated Corporate Actions for MPT Issuers** (rwa) Add a CorporateAction ledger object (owner-reserved, keylet::corporateAction(issuanceID, sequence)) created by an MPTokenIssuance's issuer via a new M

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-25 04:18	09-25 04:18	xrpl-validator	`458c2c7`	fix(node): validation relay answers the hubs' keep-alive pings
09-25 04:40	09-25 04:40	xrpl-validator	`1c373c9`	fix(node): voting-ledger validations keep their fields in canonical order
09-25 05:10	09-25 05:10	xrpl-validator	`166f033`	fix(ledger): F401 — AMM withdrawal reserve check counts XRP already paid out (soak #29 receipt)
09-25 14:07	09-25 14:07	xrpl-validator	`e63a7d9`	fix(ledger): F402 — TicketCreate paid with a Ticket does not count the ticket it spends
09-25 14:08	09-25 14:08	xrpl-validator	`0c76af1`	feat(node): stop voting for SingleAssetVault and LendingProtocol
09-25 14:31	09-25 14:31	xrpl-validator	`677c18d`	test(node): campaign 25 — Batch x Clawback, AMMClawback and token escrow (devnet vectors)

Times in UTC. A commit appears on the day it was committed to main or merged into it.